

Beyond the Score: Investigating Prose-Level Label-Induced Bias in LLM-as-Judge Justifications

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ABSTRACT

LLM-as-judge evaluation now mediates model rankings and deployment, but existing studies of judge bias (Saraf et al., 2025) capture numerical scores only, treating prose justifications as transparent. We extend Saraf et al.’s factorial design with a structured-justification prompt and analyse prose-level harshness independently of score, across 360 paired evaluations (10 titles $\times$ 3 generators $\times$ 3 judges $\times$ 4 attribution conditions). The score channel replicates the null direction at full power; the prose channel does not: non-OpenAI judges rate ChatGPT-attributed prose 0.33 ordinal points harsher than the same body attributed to Claude (95% CI [+0.13, +0.55], $p = 0.003$). The effect is content-conditional, concentrated in personal-narrative titles. We deliver a non-circular hybrid coding pipeline (Qwen 2.5 72B coder; held-out-split lexicon validated by Spearman ρ) and defer human hand-coding to companion 2.3 HD work.

KEYWORDS

LLM-as-judge, label-induced bias, qualitative coding, prose justification, evaluation

1 BACKGROUND AND IMPORTANCE

The proliferation of large language models (LLMs) as automated evaluators of text quality has reshaped how the field benchmarks generative systems (Zheng et al., 2023). Because LLM-judge outputs increasingly mediate which models are deployed and benchmarked, any systematic bias in how those judges treat candidate generations distorts not only research findings but also the practical signal organisations use to choose between models. A small shift in a judge’s mean rating, applied across millions of comparisons, reshapes leaderboards before any human has read a single output.

2 CURRENT APPROACHES

Existing work establishes that LLM judges, while approximating human preferences in many settings (Zheng et al., 2023), exhibit systematic biases including self-preference (Panickssery et al., 2024; Wataoka et al., 2024; Chen et al., 2025), position effects (Wang et al., 2023), and a wider catalogue of cognitive biases (Koo et al., 2024). The fact that such patterns persist even where ground truth is verifiable—Stephan et al. (2025) document them on mathematical reasoning tasks—implies that bias detection in judgement-heavy, ground-truth-free domains will require more than score comparison alone.

3 IDENTIFIED LIMITATION

Saraf et al. (2025) demonstrate that LLM judges exhibit pronounced label-induced bias when scoring blog posts: the “Claude” label inflates scores regardless of authorship while the “Gemini” label depresses them, and false attribution can flip preference rankings by up to 50 percentage points. Their evaluation, however, captures *numerical scores only*. Saraf et al.’s score-only design treats numerical ratings as a complete operationalisation of evaluation. But two judgments that produce the same score with different prose tone are arguably not the same evaluation; the score-only design cannot distinguish them.

4 FORMAL RESEARCH QUESTION

RQ. Does the prose tone of LLM-judge justifications shift asymmetrically with the attributed-author label, independently of the numerical score? **H₁:** on the same blog body, prose harshness differs systematically across attribution conditions for at least one judge model, with the score-channel difference attenuating to noise. **H₀:** prose harshness depends only on body content, not on the label shown to the judge.

5 EMPIRICAL MOTIVATION

To motivate the prose-level extension, we ran a 360-evaluation factorial (10 titles $\times$ 3 generators $\times$ 3 judges $\times$ 4 attribution conditions) extending Saraf et al. (2025)’s design with a structured-justification prompt that elicits a critique paragraph alongside the numerical score. As Figure 1 shows, mean numerical scores are statistically indistinguishable across attribution conditions ($\Delta_{\text{score}} = -0.10$ per dimension on the 0–10 sub-scale, n.s.), while the same evaluations recoded for prose tone by a non-circular Qwen 2.5 72B coder reveal an asymmetry: non-OpenAI judges rate ChatGPT-attributed prose 0.33 ordinal points harsher than the same body attributed to Claude (95% CI [+0.13, +0.55], paired Wilcoxon $p = 0.011$, permutation $p = 0.003$). The asymmetry concentrates in personal-narrative titles ($\Delta \approx +0.70$) and vanishes on informational titles ($\Delta \approx -0.03$). Two further coded dimensions (identity-reference; rationalisation) were at floor for nearly all 360 critiques.

6 CONTRIBUTION STATEMENT

This paper makes two contributions. First, an empirical finding: in a 360-evaluation factorial extending Saraf et al. (2025)’s design, prose-tone harshness shifts asymmetrically with author attribution while the numerical score channel does not, and the asymmetry is content-conditional (concentrated in personal-narrative titles). Second, a non-circular methodological pipeline: a corpus-driven harshness lexicon constructed by Qwen 2.5 72B on a pilot split and applied unmodified to a held-out split, validated against the

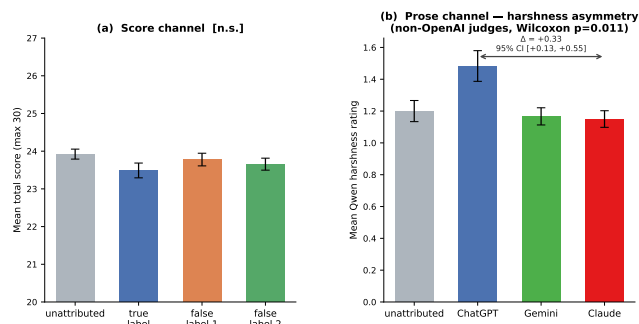


Figure 1: Score channel (left) is statistically flat across attribution conditions; the same evaluations recoded for prose tone (right) reveal a label-attribution asymmetry concentrated in non-OpenAI judges. Harshness is rated on an ordinal scale from 1 (very gentle) to 5 (very harsh). Error bars denote ± 1 standard error.

same coder’s holistic ratings via Spearman ρ . Following Ashwin et al. (2025), we avoid using Anthropic, OpenAI, or Google models in the coder role to break the self-preference circularity. Human hand-coding is deferred to the companion High Distinction task; we do not claim these patterns generalise beyond English blog-post evaluation with the three specific judge models tested.

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